

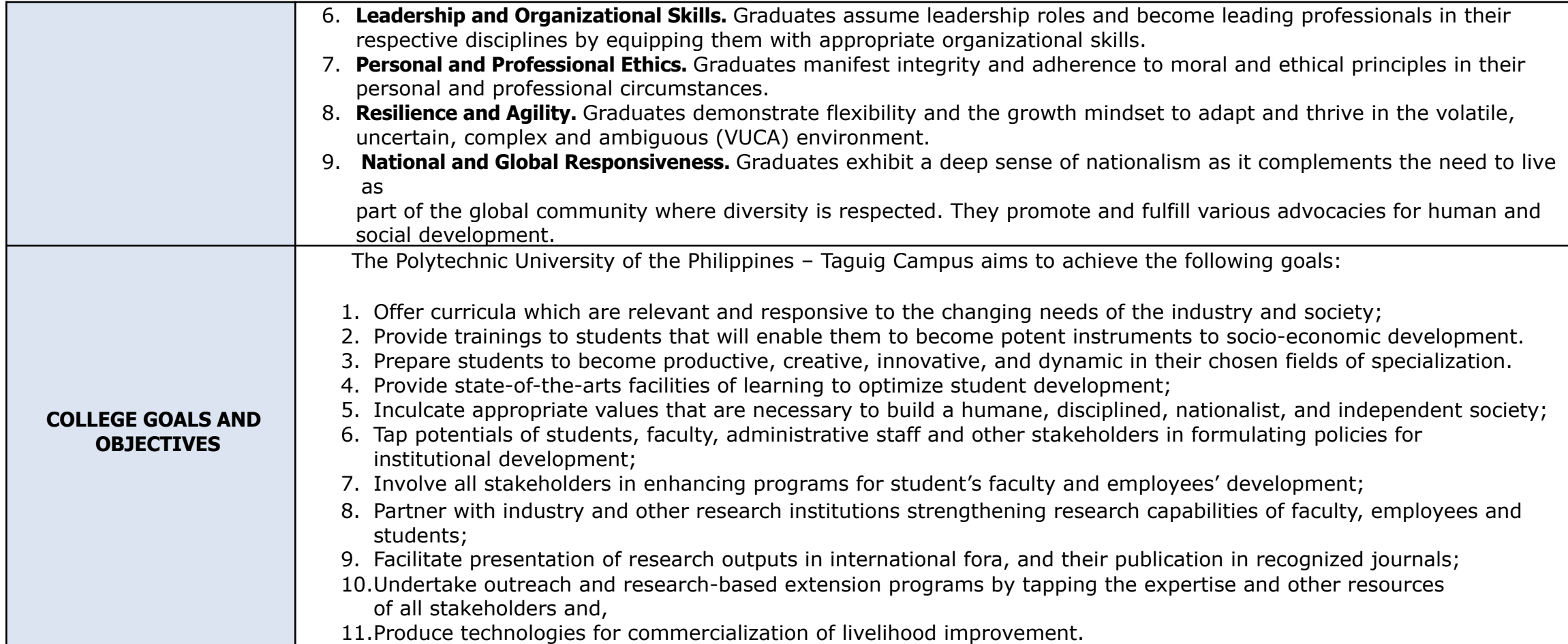


COURSE SYLLABUS						
Cours e Code	COMP 002	Course Title	Computer Programming 1		Course Credit	3
Course Description		This course introduces students to the fundamental principles of computer programming using the C++ language. It covers problem analysis, algorithm development, and structured programming concepts such as variables, data types, control structures, functions, arrays, pointers, and file handling. The course also introduces modern programming topics including concurrency and semaphores to build awareness of parallel processing and synchronization. Through hands-on coding exercises and performance-based assessments, students develop logical thinking, debugging skills, and the ability to design efficient, maintainable programs aligned with real-world applications.				
Pre-Requisites			Co-Requisites	None		
VISION		A Leading Comprehensive Polytechnic University in Asia				
MISSION		Advance an inclusive, equitable, and globally relevant polytechnic education towards national development.				
QUALITY POLICY STATEMENT		The Polytechnic University of the Philippines commits to provide inclusive and equitable quality education and promote lifelong learning opportunities for human advancement and social transformation through re-engineered polytechnic academic programs. Toward this end, we, the members of the PUP Community will vigorously and steadfastly endeavor to continuously improve the standard of University services to the satisfaction of our clients through the adoption and continuous review of our Quality Management System.				



**INSTITUTIONAL
LEARNING OUTCOMES
(ILO)**

- As a polytechnic state university, PUP shall develop its students to possess:
1. **Critical and Creative Thinking.** Graduates use their rational and reflective thinking as well as innovative abilities to life situations in order to push boundaries, realize possibilities, and deepen their interdisciplinary, multidisciplinary, and/or transdisciplinary understanding of the world.
 2. **Effective Communication.** Graduates apply the four macro skills in communication (reading, writing, listening, and speaking), through conventional and digital means, and are able to use these skills in solving problems, making decisions, and articulating thoughts when engaging with people in various circumstances.
 3. **Strong Service Orientation.** Graduates exemplify strong commitment to service excellence for the people, the clientele, industry, and other sectors.
 4. **Adept and Responsible Use or Development of Technology.** Graduates demonstrate optimized and responsible use of state- of-the-art technologies of their profession. They possess digital learning abilities, including technical, numerical, and/or technopreneurial skills.
 5. **Passion for Lifelong Learning.** Graduates perform and function in society by taking responsibility in their quest for further improvement through lifelong learning.



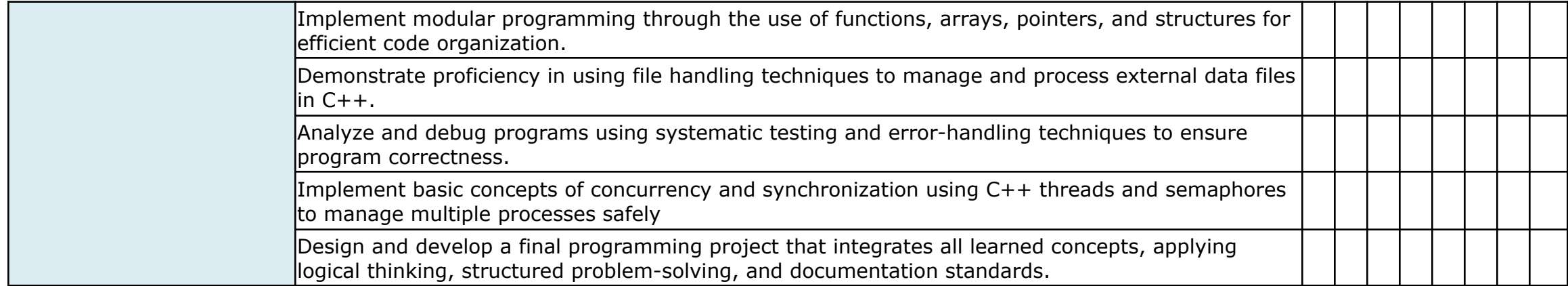
PROGRAM LEARNING OUTCOMES (PLO)	Based on CMO	Alignment to ILOs								
		1	2	3	4	5	6	7	8	9
	Apply knowledge of computing, science, and mathematics appropriate to the discipline.									

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	Analyze complex problems, and identify and define the computing requirements appropriate to its solution												
	Identify and analyze user needs and take them into account in the selection, creation, evaluation and administration of computer-based systems.												
	Design, implement, and evaluate computer-based systems, processes, components, or programs to meet desired needs and requirements under various constraints.												



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OUTCOMES-BASED TEACHING AND LEARNING PLAN (OBTL PLAN)

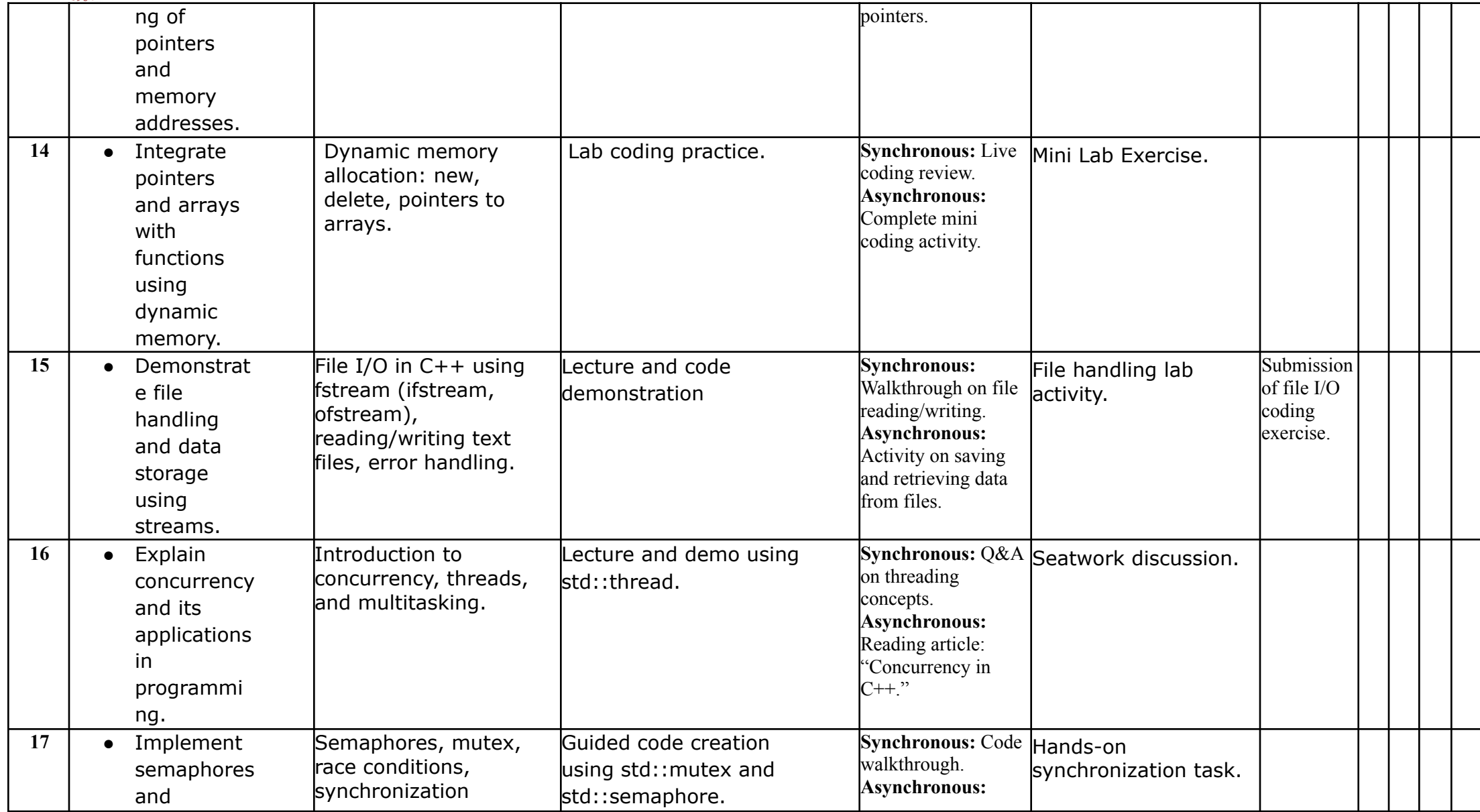
Week	Desired Learning Outcomes (DLOs)	Learning Content/ Topics	Instructional Delivery Design		Assessment		Alignment to CLOs			
			Face-to-Face	Flexible Teaching and Learning Activities (FLTAs)			1	2	3	4
					Face-to-Face	FLTAs				
1	- Equip students on the pivotal ideas embedded in the vision, mission, and 10 Pillars of PUP - Gain an overview of the course subject and requirements	- Orientation about PUP Vision, Mission, and 10 Pillars - Overview about the Subject Course and Course Requirements	Present the VMGO, policies, and guidelines on the conduct of Face-to-Face; and of the requirements of the course.		Formative Assessment (Sharing of Learning)		X			
2	Explain fundamental programming concepts and development processes.	Introduction to Programming Concepts; Steps in Program Development; Algorithms, Pseudocode, and Flowcharting.	Lecture, demo, and flowchart creation.	Synchronous: Instructor demo via IDE. Asynchronous: Watch the video on algorithm design, practice the flowchart.	Flowchart exercise submission.	Algorithm and pseudocode activity submission.				
3	Apply C++ syntax, variables, and constants to create basic	C++ structure, data types, variables, constants, and input/output.	Guided coding in IDE.	Synchronous: Q&A coding session. Asynchronous: Independent practice and reading tasks.	Coding activity in lab.	Online quiz and reflection on coding experience.				



	programs.									
4	Use operators and expressions to perform computations	Arithmetic, relational, logical operators, and precedence rules.	Code demonstration and exercises.	Synchronous: Live operator drills. Asynchronous: Problem set on expressions.	Lab performance check.	Self-check worksheet submission.				
5	Develop decision-making programs using selection structures.	if, if-else, switch statements.	Lecture and examples.	Synchronous: Problem-solving challenge. Asynchronous: Study resource on conditionals (W3Schools).	Code-writing quiz					
6	Apply nested and multiple conditionals for complex logic.	Nested if, ternary operator, and multi-way selections.	Guided coding and problem-solving.	Synchronous: Interactive exercise. Asynchronous: Coding challenge.	Short lab quiz					
7	Apply iteration structures for repetitive processes.	for, while, do-while loops; break and continue.	Demo and loop exercises.	Synchronous: Live code tracing. Asynchronous: Practice loops with online IDE.	Lab hands-on performance.					

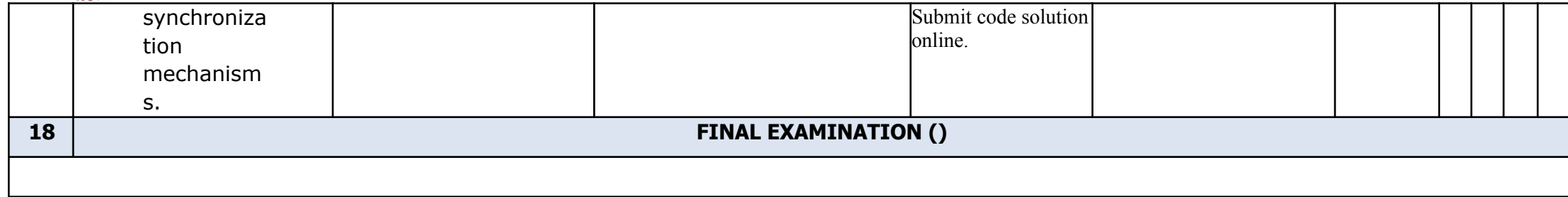


8	- Combine conditionals and loops to solve problems. - Create modular programs using functions.	- Nested loops and counter-controlled repetition. - Function declaration, parameters, and return types.	Coding workshop. Lecture, guided lab demo.	Synchronous: Problem-solving collaboration. Asynchronous: Drill worksheet.	Function-based coding task.	Function tracing activity.				
9	MIDTERM WRITTEN EXAMINATION									
10	Implement recursion in problem-solving.	Recursive functions, base and recursive cases.	Tracing recursive examples.	Synchronous: Discussion and feedback. Asynchronous: Practice recursive code.	Lab evaluation on recursion.	Short written reflection on recursion logic.				
11	Use arrays to process multiple data items.	1D arrays: declaration, initialization, and traversal.	Lab coding and demonstration.	Synchronous: Code review. Asynchronous: Array exercises online.	Hands-on lab quiz					
12	Apply 2D arrays for matrix operations.	2D arrays and nested loop operations.	Demo and collaborative exercise.	Synchronous: Group coding via breakout rooms. Asynchronous: Practice set on 2D arrays.	Practical test on arrays.	Peer discussion and Q&A board.				
13	Demonstrate understanding	Pointer declaration, dereferencing, and pointer arithmetic.	Lecture and tracing session.	Synchronous: Example analysis. Asynchronous: Watch tutorial on	Coding exercise (pointer tracing).					



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		Resources and Applying Strategies in Teaching Prose <i>Competency 4:</i> Recognizing Assessment in Teaching Prose	achievement which are consistent with the selected competencies • Learning Plan Making according to the English curricula that develops higher order thinking skills of learners through the use of prose and dramatic texts		• Quizzes					
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18

FINAL EXAMINATION

LIST OF REFERENCES

BOOKS:

WEBSITE:

- GeeksforGeeks. (2025, July 23). *Concurrency in C++*. GeeksforGeeks. <https://www.geeksforgeeks.org/cpp-concurrency/> GeeksforGeeks
- GeeksforGeeks. (2025, October 8). *File Handling in C++ Classes*. GeeksforGeeks. <https://www.geeksforgeeks.org/file-handling-c-classes/GeeksforGeeks+1>
- Li, D. (2023, September 17). *C++ Multithreading and Concurrency: A Comprehensive Guide*. FriendlyUsers Tech Blog. [https://friendlyuser.github.io/posts/tech/2023/C%2B%2B Multithreading and Concurrency A Comprehensive Guide/](https://friendlyuser.github.io/posts/tech/2023/C%2B%2B+Multithreading+and+Concurrency+A+Comprehensive+Guide/) friendlyuser.github.io
- Grimm, R. (2023). *Modern C++ Training/Workshop: Concurrency in Modern C++*. (Training material). ModernesCpp. <https://www.modernescpp.org/wp-content/uploads/2023/04/Concurrency.pdf>
- W3Schools. (n.d.). *C++ Pointers*. W3Schools. Retrieved October 14, 2025, from https://www.w3schools.com/cpp/cpp_pointers.asp W3Schools
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- W3Schools. (n.d.). C++ Functions. W3Schools. Retrieved October 14, 2025, from https://www.w3schools.com/cpp/cpp_functions.asp W3Schools
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- GeeksforGeeks. (2023, September 22). C++20 <semaphore> Header. GeeksforGeeks. Retrieved October 14, 2025, from <https://www.geeksforgeeks.org/cpp/cpp-20-semaphore-header/> GeeksforGeeks
- CodeWithC. (2023, October 27). C++ Concurrency: Understanding Semaphore Mechanism. CodeWithC. Retrieved October 14, 2025, from <https://www.codewithc.com/c-concurrency-understanding-semaphore-mechanism/> codewithc.com
- Medium. (2025, July 19). Semaphores in C++. Medium. Retrieved October 14, 2025, from <https://medium.com/%40kamresh485/semaphores-in-c-b0e98ee95436> Medium
- Medium. (n.d.). Unlocking Concurrency: The Guide to Locks, Semaphores, and Mutexes. Medium. Retrieved October 14, 2025, from <https://medium.com/%40sumit-s/unlocking-concurrency-the-guide-to-locks-semaphores-and-mutexes-8f5a579d7cb>

CLASSROOM POLICY	
FACE-TO-FACE DELIVERY	FLEXIBLE TEACHING AND LEARNING ACTIVITIES (FLTAs)



Please refer to the FLTA policies, except for the stipulation on Attendance. Attendance will be checked during face-to-face sessions.

Students' Responsibilities

Students are expected to:

- Take time to read and watch the provided class learning materials.
- Fulfill class requirements such as posting on discussion boards and submitting group presentations.
- Check the status of their submissions and inform the instructor if they are having problems submitting the requirements.



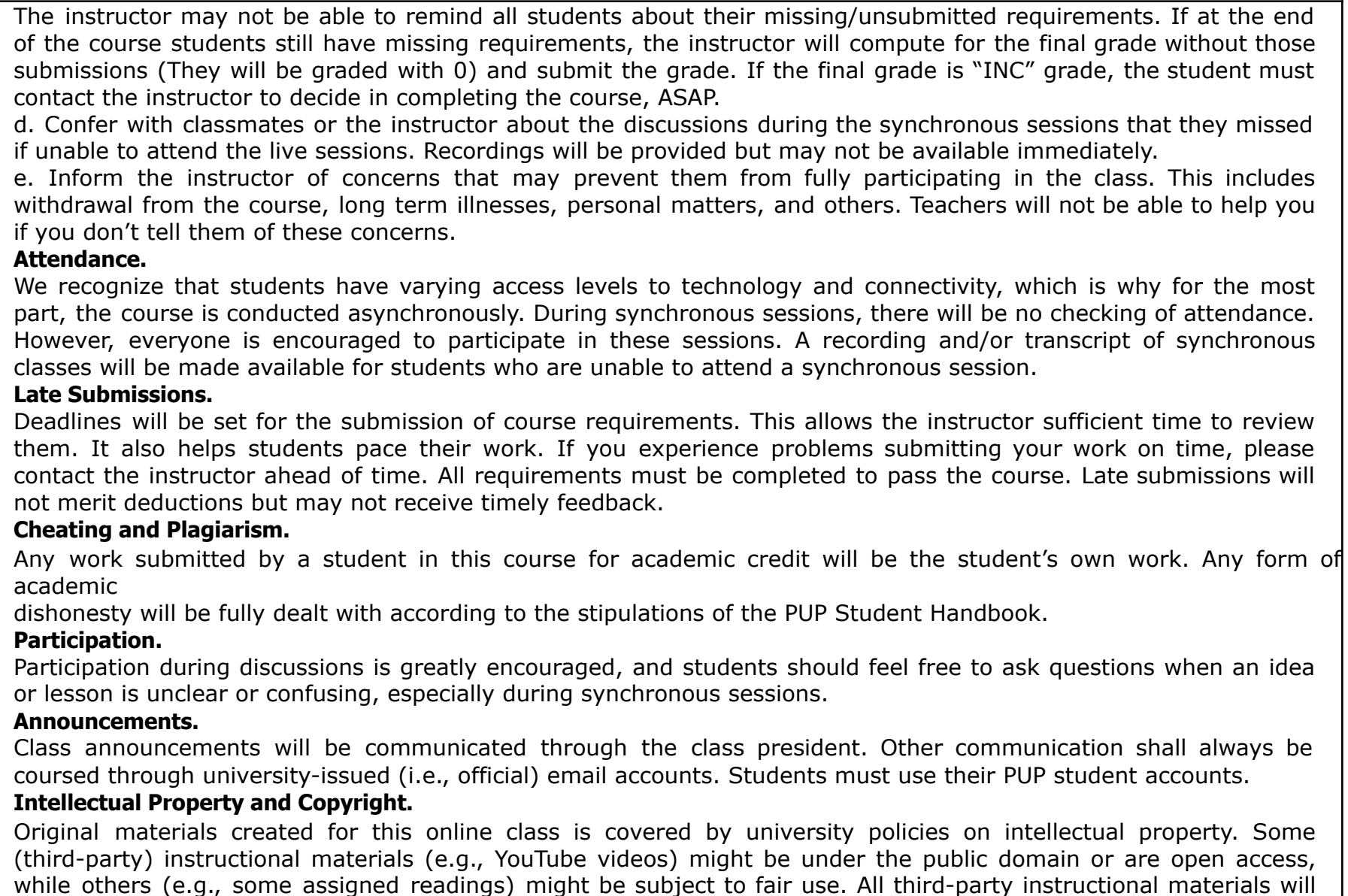
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OFFICE OF THE VICE PRESIDENT FOR CAMPUSES
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(BSIT)



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	be attributed to their original creators. Electronic copies of the assigned readings are for your private consumption only. Access to the course materials for this class for PUP students will be restricted to only those students registered in this course and will be available for only a
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2025
7th in the Philippines
92nd in Southeast Asia
541-560 in Asia



The WORLD
UNIVERSITY
RANKINGS
for INNOVATION

Rank 21
GENERATIVE AI

Rank 39
STUDENT SUPPORT

Rank 50
SYMBOL/



2025
4★
5★
6★
7★
8★
9★
10★



Times Higher Education
Impact Rankings 2024
DEBUTS AT 1501+

Employability

Teaching

Research

Internationalization

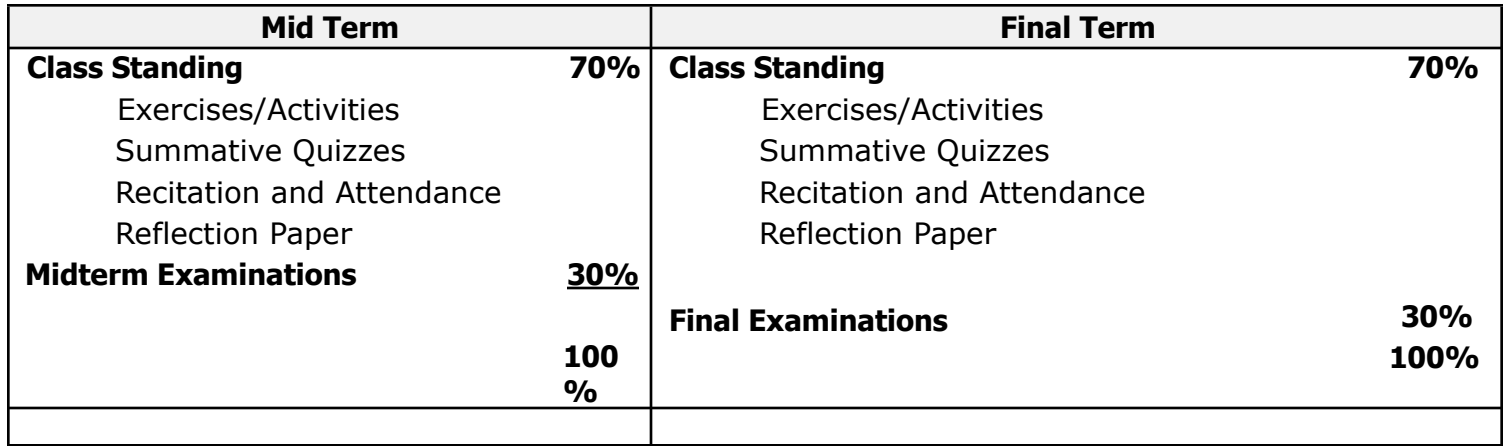
Online Learning

Academic Development

Inclusiveness

Social Responsibility

Specialist Criteria: Bachelor of Science in Nutrition and Dietetics (BSND)



	limited period. Please do not share these materials with anyone elsewhere. The same applies to the recordings of the synchronous sessions.
COURSE REQUIREMENT/S	GRADING SYSTEM
- Students are required to complete hands-on programming exercises, quizzes, and laboratory activities to demonstrate understanding of C++ programming concepts. - They must also submit algorithms, flowcharts, and machine problems that integrate course topics such as loops, functions, arrays, and file handling.	Final Grade = (Midterm Grade + Final Term Grade) / 2

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CLASS INFORMATION	FACULTY INFORMATION
Course, Year and Section: Computer Programming, BSIT 1-1	Name of Faculty: ALMIRANEZ, GECILIE C.
Day and Time: T - 07:30PM - 09:00PM W - 07:30PM - 09:00PM TH - 07:30PM - 09:00PM F - 07:30PM - 08:00PM	Consultation Time:
Room: DOST LAB	Office Tel. No./ Mobile Phone No.
School Year: 2025-2026 Semester: First Semester	E-Mail Address:

Prepared by:	Reviewed and checked by:	Recommending Approval:	Approved by:
ALMIRANEZ, GECILIE C. Faculty Member	MS. DIANNE S. SEGURORA Head of Academic Programs	DR. RHYAN V. MOLINAR Campus Director	DR. ARMANDO A. TORRES Vice President for Campuses